

QEBreak at WMT26: A Pre-Registered Audit of Omission and Addition Asymmetry in Machine Translation Quality Metrics

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Abstract

Reference-free quality estimation (QE) metrics see only the source and the hypothesis, so when a translation silently drops source content there is nothing in the hypothesis for them to flag. We audit this structural blind spot with QEBreak, a pre-registered contrastive challenge set submitted to the WMT26 Automated Translation Quality Evaluation task (Subtask 4, Challenge Sets): 2,886 segments over 10 language directions, organized as matched {base, omission, addition} triples, mostly on real WMT24 system outputs (Arm B, the headline contrast), a fluent fact-free verbosity dose ladder (Arm A), and numeric-contradiction minimal pairs as a positive control (Arm C). All hypotheses, endpoints, equivalence margins, strata, and the orientation manifest were frozen in git-timestamped commits before any organizer score existed. The returned scores from 31 systems confirm the deficit and refute our proposed mechanism. Reference-free QE metrics under-penalize omission (family asymmetry $A = -0.51$, 95% CI $[-0.57, -0.44]$; omission-over-base preference flips reach 19.5% for the xCOMET-XL QE baseline), and the direction by reference-availability interaction is significant ($\beta = -0.069$, CI $[-0.081, -0.057]$); yet the reference-based neural family is itself far from symmetric ($A = -0.42$, CI $[-0.49, -0.35]$, entirely outside the pre-registered ± 0.30 equivalence margin), so the pre-registered anchor fires its refutation leg: omission under-penalization is a pervasive property of learned metrics, one that reference access softens but does not remove. The pre-declared provenance gradient confirms that subtle LLM-generated edits on real system outputs cut deepest ($A = -0.60$ in the powered stratum), and the effect survives both anti-circularity confirmation strata. We describe the set’s construction, its validity gates, the blind submission protocol, and the pre-registered analysis that let our own headline

hypothesis fail in public while the measurement it produced stands.

1 Introduction

A quality estimation metric that scores translations without a reference is the natural reward signal for machine translation systems: it is cheap, it needs no human reference, and it is already used to rerank candidate outputs. That role raises a structural question. A reference-based metric can notice that translated content is missing because the reference still contains it. A reference-free metric has only the source and the hypothesis, and a hypothesis that omits content is shorter, more fluent, and internally consistent. If reference-free metrics penalize omission less than they penalize a comparable amount of added content, then a best-of- n reranker built on such a metric will sometimes prefer the incomplete translation. QEBreak was built to measure that preference directly.

QEBreak tests three pre-registered hypotheses. First, the headline hypothesis H_B : reference-free QE metrics under-penalize meaning omission relative to fluent addition, while reference-based metrics penalize the two directions roughly symmetrically. Second, H_A : reference-free QE scores do not decrease, or even increase, as fluent, fact-free, source-unsupported material is appended to a realistic base translation. Third, H_C , a positive control: the reference-based neural frontier catches a single load-bearing numeric contradiction while surface and reference-free QE metrics diverge, which proves the harness can detect breaks at all.

The design choice that separates this audit from prior challenge sets is that every claim was frozen before any score existed. The hypotheses, the primary endpoints, the equivalence margin for the reference-based anchor, the multiplicity correction, the provenance strata, and the metric orientation manifest were committed to a timestamped git history before the challenge set was uploaded, and

the three pre-score amendments are themselves logged commits (Section 4). Whatever the organizer scores say, the reported outcome is the pre-registered one: a confirmation, or an honest negative.

Our contributions are: (1) a 2,886-segment contrastive challenge set over 10 language directions, built as matched same-source triples with controlled information-unit edits, submitted blind with opaque identifiers and a private keymap; (2) a pre-registered analysis with a signed interaction primary endpoint, a reward preference-flip rate that directly measures the reranking failure, and a numeric equivalence anchor for the reference-based family; (3) the findings themselves: a significant reference-availability interaction on top of an asymmetry that turns out to be shared by every learned family, a preference-flip rate that reaches one matched contest in five for a current QE baseline, a verbosity hypothesis refuted outright, and two returned systems that score corrupted numerals above correct ones.

2 Related Work

ACES (Amrhein et al., 2022) and its span-annotated extension (Moghe et al., 2025) established the contrastive challenge-set paradigm, including single-edit omission and addition items. QEBreak differs on three axes: the omission and the addition are matched on the same source at a controlled information-unit dose, the contrast is carried through the reference-availability axis, and the analysis reaches the reward-model preference flip that measures the deployment-relevant event directly. The MT-breaker line of adversarially difficult inputs (Zouhar et al., 2026) sets the rigor bar we adopt. On the length axis, prior work diagnosed QE length bias (Zhang et al., 2025) and a WMT25 challenge set measured continuation drift (CoDrift, in Lavie et al., 2025); Arm A instead isolates pure, fact-free, unsupported length as a dose curve. The pre-registered family taxonomy covers surface metrics (Papineni et al., 2002; Popović, 2015; Snover et al., 2006), reference-based neural metrics (Rei et al., 2022a; Sellam et al., 2020; Guerreiro et al., 2024; Juraska et al., 2024), reference-free QE (Rei et al., 2022b), and LLM judges in the GEMBA family (Kocmi and Federmann, 2023).

Direction	A	B	C	Total
en-de	330	300	264	894
en-cs	0	219	210	429
cs-uk	0	261	156	417
en-ru	126	45	192	363
en-zh	0	66	188	254
cs-de	0	0	216	216
en-ar	0	0	176	176
en-ko	0	0	66	66
en-ja	0	33	18	51
zh-ja	0	0	20	20
Total	456	924	1,506	2,886

Table 1: Shipped segments per language direction and arm. Arm A: 76 six-step verbosity dose series (456 segments). Arm B: 308 matched {base, omission, addition} triples (924 segments). Arm C: 753 numeric minimal pairs (1,506 segments).

3 The QEBreak Challenge Set

QEBreak was submitted to WMT26 Subtask 4: 2,886 segments across 10 language directions, 231,582 tokens under the organizer counting rule, well under the 1M-token cap; the organizers confirmed receipt on 22 July 2026. Table 1 gives the composition.

3.1 Three arms

Arm B (headline): matched omission and addition triples. Each of the 308 triples holds the source fixed and pairs a base translation with an omission that removes one information unit and an addition that appends one fact-free, source-unsupported unit. Members travel as separate rows paired by opaque group identifiers, matching the organizer-endorsed separate-segments design. The triple supports both primary endpoints: the signed direction-by-reference-availability interaction (N1) and the reward preference-flip rate P_{flip} , the fraction of matched {base, omission} pairs where a reference-free metric scores the omission at or above the base (N2). The directions are matched on information units, not edit size: additions change a median of 2.0 times as many characters as omissions remove, so a length-controlled sensitivity refit accompanies the primary interaction (Section 4).

Arm A (secondary): a verbosity dose ladder. Each of the 76 series appends 1 to 5 fluent, fact-free filler units to a base translation on en-de and en-ru, giving six segments per series including the dose-0 base. Fillers are screened to contain no digits, dates, or named entities, and to be entailment-

neutral against both the source and the base (Section 3.2), so the ladder isolates pure unsupported length. At the maximum dose the hypothesis grows to a median of 6.4 times the base length.

Arm C (positive control): numeric minimal pairs. Each of the 753 pairs contrasts a correct hypothesis with a copy whose single load-bearing numeral is corrupted: digit substitution (440 pairs), transposition (213), or a magnitude shift (100). Numeral selection follows a scoring rule frozen in the pre-registration (units and separators weighted up, years penalized and never corrupted as a fallback), so the corrupted numeral is never a researcher degree of freedom. Arm C spans all 10 directions and proves the harness has teeth: if the reference-based frontier misses a flat numeric contradiction, the harness, not the metrics, is under suspicion.

3.2 Validity gates

Every generated item passes deterministic validity gates before it can ship; items that fail are dropped, not repaired. The omission gate requires the reference to entail the omitted-member hypothesis while the dropped unit stays un-entailed by the remaining text (mDeBERTa-v3-base-xnli, [Laurer et al., 2024](#)), which blocks omissions that merely rephrase. The addition gate requires the appended unit to carry no digit, date, or named entity (XLM-R NER) and to be entailment-neutral against both source and base (neutral ≥ 0.70 , entailment and contradiction ≤ 0.15 each), validating the true appended delta rather than the generator’s self-report. Subtle LLM-generated triples obey symmetric length caps (at most 25 percent dropped or grown; realized worst cases 22.7 and 23.7 percent), and all perturbed members must be distinct from their base and from each other.

Gate survival is low by design: between 2.6 and 10 percent of raw candidate pairs survived all gates in the logged powered runs (402 candidates yielded 40 gated triples on cs-uk; 154 yielded 4 on en-ja).

3.3 Provenance strata

Arm B ships in three pre-declared strata, defined purely from metadata written at generation time. S1 (89 triples): subtle perturbations generated by Llama-3.1-8B on real WMT24 system outputs under the length caps; the omission typically drops a single detail such as a street num-

ber. S2 (100 triples): deterministic whole-unit edits on real WMT24 system-output bases. S3 (119 triples): deterministic edits on reference-verbatim bases. Real-MT bases come from the median-ranked WMT24 system per direction ([Kocmi et al., 2024](#)), watermark canary lines filtered out. The pre-registered secondary prediction is a gradient: $A(S1) < A(S2 + S3)$ and $P_{\text{flip}}(S1) > P_{\text{flip}}(S2 + S3)$, no prediction between S2 and S3, and no stratified test in the confirmatory family. The rationale was recorded before any score existed: the local CometKiwi proxy already penalizes the crude deterministic omissions, so pooled dilution is a described property of the design, not a post-hoc rescue.

3.4 Blind submission protocol

The uploaded rows carry exactly the seven published fields; all internal metadata was stripped, identifiers are opaque hashes with no label, phenomenon, or dose token, and row order was decorrelated so the base member sits at no fixed position (base-first fell from 100 percent to 44.0 percent, against a 43.3 percent chance level). A private keymap, never uploaded, restores the pairing when scores return. Residual structure inherent to contrastive sets, such as pair members sharing a source, is disclosed in Section 7.

4 Pre-Registered Analysis Plan

The full pre-registration lives in a timestamped git history frozen before the challenge-set upload and before any organizer score existed: the registration (commit 6febc4b, 9 July), a sign-gloss erratum (b8e78f3), the scope-and-anchor amendment (91c19c1), and a teeth-wording clarification (98bf78c), all predating the 16 July upload; the orientation manifest is frozen by content hash. After the return arrived, the only changes were the ones the registration itself permits: additive orientation declarations for the returned roster, and the axis scoping of the twelve undocumented systems, both committed with a decision-log entry before any score value was analysed. Format inspection of the return had unavoidably displayed a handful of raw rows from two files before that commit; no aggregate was computed and no declaration used score values, as logged (Section 5). The build pipeline, the frozen analysis code, the labeled challenge set, and the scoring keymap are available from the author on request; the camera-

ready will carry the archive decision once the organizers identify each team’s main submissions.

Primary endpoints. N1 is the signed interaction coefficient β (direction $\times$ reference-availability) from one mixed-effects model per arm on rank-normalized signed score drops (rank-normalization is the pre-registered primary transform; a z-score refit is the declared sensitivity analysis), with inference from source-clustered bootstrap (B=2000) 95 percent CIs as the sole load-bearing procedure. N2 is P_{flip} for reference-free QE metrics. Arm A’s endpoint is the within-source Spearman ρ between dose and oriented score ($H_0: \rho \geq 0$). Arm C’s endpoint is Kendall τ_b ($H_0: \tau_b \leq 0$) with the tie rate reported as a first-class number. One addition post-dates the frozen registration but precedes any score (declared 4 August, while scores were still pending): a sensitivity refit of the Arm B interaction with edit magnitude (absolute character change) as a covariate, motivated by the measured 2.0x median edit-size imbalance between additions and omissions; by the frozen family rule it is exploratory and can only qualify, never rescue, the primary result.

The reference-based anchor. H_B is a contrast, so it needs the reference-based family to actually be symmetric. The pre-registered anchor defines a scale-free relative asymmetry $A_m = (|b_{om,m}| - |b_{add,m}|) / ((|b_{om,m}| + |b_{add,m}|) / 2)$ per metric, bounded in $[-2, 2]$, with $A_m := 0$ when both drops are exactly zero; family-level A is the unweighted mean of A_m over that family’s returned roster. The anchor is established only if the 90 percent bootstrap CI of $A_{\text{ref-based}}$ lies inside the equivalence margin $[-0.30, +0.30]$ (TOST logic), plus a teeth precondition: the pooled reference-based mean drop, both directions pooled, must have a 95 percent bootstrap CI entirely above 0, since a family that barely penalizes either edit cannot anchor anything. If the CI is not inside the margin the anchor is not established and is reported with its CI; that alone neither confirms nor refutes H_B . The predicted signs, consistent with the logged pre-score erratum on the θ gloss, are $A < 0$ for reference-free QE (omission under-penalized) and A inside the margin for reference-based neural metrics.

Multiplicity and orientation. The confirmatory family is exactly one test per metric-arm (Arm

A ρ , Arm B β within reference-free QE, Arm C τ_b), with Benjamini-Hochberg FDR at $q = 0.05$ applied once (Benjamini and Hochberg, 1995); everything else is labeled exploratory. Every returned metric whose orientation matches the frozen manifest is included mechanically, and the family size m is computed from that roster and logged before any score value is inspected. The harness fails closed on any undeclared metric name; the only permitted post-return change is adding a pattern for a returned name, in a descendant commit made before any score value is inspected and carrying its own decision-log entry. A pre-declared minimum-metrics gate applies: if fewer than two reference-free QE metrics are returned, the family-level asymmetry gloss is not estimated and per-metric signed drops are reported instead.

Falsification. If the returned scores show reference-free QE approximately symmetric, or reference-based neural metrics equally asymmetric (the operationalized leg: the 95 percent CI of $A_{\text{ref-based}}$ entirely below -0.30), H_B is refuted, and this paper reports that refutation as the pre-registered outcome.

Anti-circularity. A local CometKiwi pre-filter, which belongs to the same family as the audit target, never gated item selection for Arms A and B; it only attached a prediction label, and the shipped set is balanced over all validity-gate-passing items, which is what rules out selection on the label. The label drives a pre-declared stratified split. H_B is reported as confirmed only if the effect also holds in the stratum the pre-filter predicted would not break, and on the real-WMT-system-output subset.

5 Results

The organizers returned segment scores from 31 systems on 6 August 2026 (the Task 2 protocol: a quality score per segment, higher is better). 28 systems cover all 2,886 segments; cuni-v14-regression covers three directions and the two Lexicala QE ensembles cover three of the four Arm-C-only directions (cs-de, en-ar, zh-ja). The roster is a superset: some teams submitted more than the allowed two systems, and near-duplicate submission-id variants remain until the organizers identify main submissions. Twelve systems carry opaque submission names whose refer-

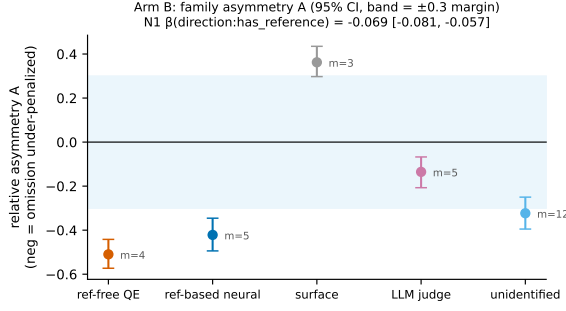


Figure 1: Family-level relative asymmetry A with clustered-bootstrap 95% CIs; the shaded band is the pre-registered ± 0.30 equivalence margin for the reference-based anchor. Ref-free QE -0.51 , ref-based neural -0.42 (outside the margin: the anchor’s refutation leg fires), unidentified participants -0.32 , LLM judge -0.14 , surface $+0.36$.

ence usage no public source documents yet; under the fail-closed rule they were declared with orientation only, in a pre-analysis commit (Section 4), so they enter every per-metric endpoint mechanically but sit out the reference-availability axis, which is undefined for them. All ingestion checks passed: every returned segment identifier matched the private keymap, and per-system counts matched the organizers’ coverage table exactly. Of the $m = 60$ confirmatory tests, 58 reject at $q < 0.001$; the two exceptions are the Arm C failures of the two MQM-LLM variants (Section 5.4).

5.1 Headline: the omission and addition asymmetry (Arm B)

The signed interaction is significant in the predicted direction: $\beta(\text{direction} \times \text{reference-availability}) = -0.069$, source-clustered bootstrap 95% CI $[-0.081, -0.057]$, $q < 0.001$ in the confirmatory family; the addition-minus-omission penalty gap is wider when the metric has no reference. The edit-magnitude sensitivity refit leaves it essentially unchanged ($\beta = -0.073$, CI $[-0.084, -0.061]$), so the 2.0x character imbalance of Section 3 does not explain it. The family asymmetries tell the larger story. Reference-free QE: $A = -0.510$ $[-0.573, -0.442]$, with per-metric A_m of -1.28 (xCOMET-XL-QE), -0.59 (xCOMET-XXL-QE), -0.22 (cuni-v14-regression), and $+0.05$ (COMETKiwi22). Reference-based neural: $A = -0.422$ $[-0.494, -0.346]$. LLM judges: $A = -0.135$ $[-0.207, -0.067]$. The

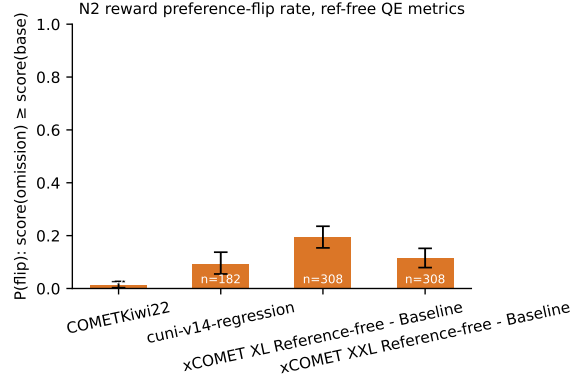


Figure 2: P_{flip} per reference-free QE metric with clustered-bootstrap 95% CIs: xCOMET-XL-QE 0.195, xCOMET-XXL-QE 0.114, cuni-v14-regression 0.093, COMETKiwi22 0.013.

twelve unidentified systems: $A = -0.323$ $[-0.395, -0.250]$. Surface metrics alone over-penalize omission: $A = +0.362$ $[+0.297, +0.435]$, as n-gram precision plus a brevity penalty punishes missing content twice. The anchor verdict is this paper’s central honest finding. The teeth precondition holds (the pooled reference-based drop has 95% CI $[+11.5, +13.4]$, comfortably above zero), but the reference-based 90% and 95% CIs lie entirely below the -0.30 margin, so the pre-registered refutation leg of Section 4 fires. H_B ’s mechanism claim is refuted: omission under-penalization is not a structural privation of reference-free scoring. It is a pervasive property of learned metrics, one that reference access reduces (the significant interaction) and that surface overlap inverts.

5.2 The reward preference flip (N2)

P_{flip} is the deployment-relevant number: the rate at which a reference-free metric scores the omission at or above its own base, which is exactly the contest a best-of- n reranker adjudicates. Over 308 matched pairs (182 for the partial-coverage CUNI system): xCOMET-XL reference-free 0.195 $[0.154, 0.235]$, xCOMET-XXL reference-free 0.114 $[0.079, 0.152]$, cuni-v14-regression 0.093 $[0.055, 0.137]$, COMETKiwi22 0.013 $[0.003, 0.027]$. A reranker guided by the xCOMET-XL QE baseline prefers the translation that silently dropped content in one matched contest out of five. COMETKiwi22 almost never flips outright and is also the family’s most symmetric member (the per-metric A_m above); the deep asymmetry is carried by the xCOMET

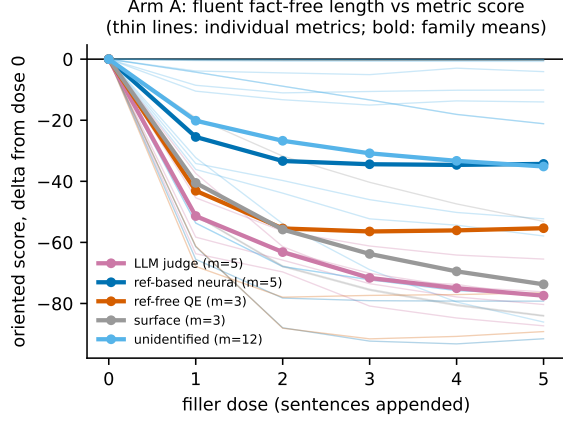


Figure 3: Oriented score against fluent fact-free filler dose (thin lines: individual metrics; bold: family means). Every family’s mean falls monotonically; no returned metric rewards pure unsupported length.

QE baselines. The failure is graded, not uniform, inside the family.

5.3 Length bias (Arm A)

H_A is refuted outright, and we report the pre-registered negative as good news for the field: every returned metric penalizes fluent fact-free filler, with within-source Spearman ρ from -0.37 (fluency2-gemini35-esa) to -1.00 (BLEU, chrF, spBLEU, YiSi-1), all $q < 0.001$ against H_0 : $\rho \geq 0$. COMETKiwi22 reaches $\rho = -0.86$ $[-0.89, -0.83]$; the xCOMET reference-free baselines -0.65 and -0.70 . The length-gaming attack this arm operationalizes does not work on the modern roster. Reference-based curves are exploratory context on this arm (reference-verbatim bases inflate the dose-0 intercept), and H_A conclusions are scoped to en-de and en-ru as registered.

5.4 Positive control (Arm C)

The control holds: the harness has teeth. The reference-based neural frontier catches the single corrupted numeral (COMET22, BERTScore, and YiSi-1 at $\tau_b = +1.00$; xCOMET-XXL reference-based $+0.95$), surface metrics detect the token change deterministically ($+1.00$), and the reference-free baselines trail visibly (COMETKiwi22 $+0.77$, xCOMET reference-free $+0.77$ and $+0.79$): reference-based catches while reference-free trails, as predicted. One cell of the frozen prediction table was wrong and we report it rather than reframe it: surface metrics were predicted to break on this arm and instead held perfectly. Contrastive accuracy here counts metric

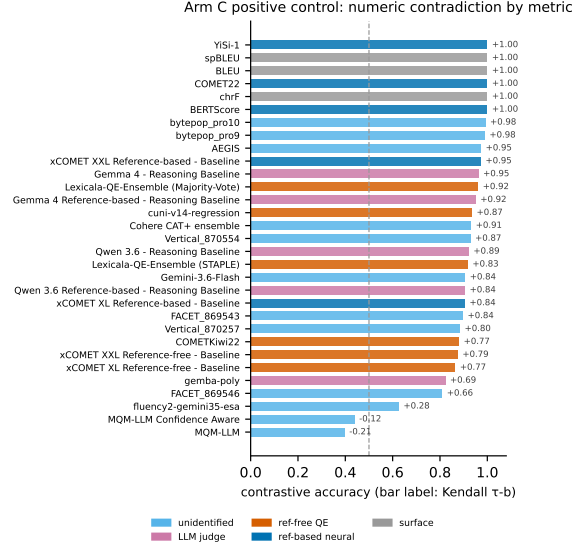


Figure 4: Numeric-contradiction contrastive accuracy, sorted; bar labels give Kendall τ_b . All families sit far above chance except the two MQM-LLM variants at the bottom, which invert the control.

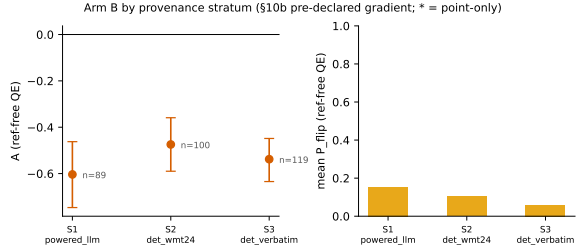


Figure 5: The pre-declared provenance-strata gradient for reference-free QE: relative asymmetry A (left) and mean P_{flip} (right). The powered subtle stratum S1 cuts deepest on both readings, as pre-registered.

ties as half credit, sharing the τ_b denominator; tie rates stay at or below 0.19 and are highest for span-derived scorers. Two participant systems invert the control: MQM-LLM ($\tau_b = -0.21$, contrastive accuracy 0.397) and MQM-LLM Confidence Aware ($-0.12, 0.439$) score the corrupted member above the correct one more often than not, the only two confirmatory non-rejections on the board. The empirical orientation cross-check flags both, and their strongly negative Arm A slopes confirm the declared orientation is right, so this is a genuine system-level failure on load-bearing numerals, not a sign error in our manifest.

5.5 Stratified reading

The pre-declared gradient confirms. In S1 (subtle LLM perturbations on real system outputs, 89 triples) the reference-free asymmetry deepens

to $A = -0.604$ and mean P_{flip} rises to 0.154, against -0.474 and 0.104 in S2 and -0.538 and 0.058 in S3; the interaction strengthens to $\beta = -0.117$ $[-0.151, -0.082]$ inside S1. The prediction $A(\text{S1}) < A(\text{S2} + \text{S3})$ and $P_{\text{flip}}(\text{S1}) > P_{\text{flip}}(\text{S2} + \text{S3})$ holds on both axes. Both anti-circularity confirmation strata of Section 4 hold as well. In the 304 pairs the local CometKiwi proxy predicted would NOT break: $\beta = -0.069$ $[-0.081, -0.057]$, $A(\text{ref-free}) = -0.503$, mean $P_{\text{flip}} = 0.094$. On the real-WMT24-system-output subset (189 pairs): $\beta = -0.078$ $[-0.094, -0.061]$, $A(\text{ref-free}) = -0.487$, mean $P_{\text{flip}} = 0.129$. The measured deficit is not an artifact of proxy-guided construction, and it lives on real MT output. Had the anchor held, these strata would have completed a confirmation of H_B ; with the anchor refuted, they instead certify that the pervasive asymmetry the audit found is real wherever we can check it.

6 Human Validation Protocol

We built a blind human validation study before submission: a 270-item single-annotation sheet plus a 60-item double-annotated overlap, sampled with a guaranteed minimum of powered-stratum items and sealed so annotators see only an identifier, the source, and the hypothesis. The protocol targets per-direction item precision of at least 95 percent and reports Cohen’s κ and Krippendorff’s α on the overlap (Cohen, 1960; Krippendorff, 2018). Annotation had not been completed at submission time, so this paper reports the protocol and no human precision numbers; the machine-side gates of Section 3.2 are the operative item-validity evidence, and the sealed sheet ships with the release so the study can be run independently of this paper’s claims.

7 Conclusion

QEBreak turns a structural suspicion about reference-free QE metrics into a falsifiable, pre-registered measurement: matched same-source omission and addition contrasts, carried through the reference-availability axis, down to the reward preference flip that a metric-guided reranker would actually commit. The returned scores confirmed the deficit and refuted our favorite explanation of it: reference-free QE metrics prefer the incomplete translation often enough to matter, but so, to a smaller degree, do the reference-based metrics that were supposed to anchor the contrast, so

the blind spot belongs to learned metrics as a class rather than to reference-freeness, and the significant interaction shows reference access narrows it without closing it. What we would hand next year’s breakers is the discipline rather than the hypothesis: the blind protocol with a private keymap, the fail-closed orientation manifest, and the equivalence anchor with its teeth precondition, which together let a wrong mechanism story die in public while the measurement that killed it stands.

Limitations

- **Circularity is disclosed, not eliminated.** CometKiwi labeled (never gated) Arm A and B items, and the confirmation rule requires the effect to survive in the stratum predicted not to break and on real system outputs. Within-family corroboration is never treated as independent transfer.
- **Contrastive structure is partly visible.** Members of a pair share a source, group size correlates with arm, and control members are reference-verbatim. These properties are inherent to contrastive minimal-pair sets; an honest metric scores each row independently and cannot exploit them, but a deliberately adversarial scorer could. The blind protocol removes the fixable tells (labels, id tokens, positional order); the inherent ones are stated here.
- **Single-dose Arm B.** Every shipped omission and addition carries one information unit; the multi-dose ladder was not realized (8 GB single-machine generation), so no dose-interaction claim is made and the reduced-design interaction is the primary endpoint.
- **Arm A scope.** 76 series on two directions with reference-verbatim bases. For reference-free QE, the target family, base equals reference is mechanically invisible; for reference-based metrics the dose-0 intercept is inflated, so that curve is exploratory context only. H_A conclusions are pre-registered as scoped to these two directions (en-de, en-ru).
- **Arm B coverage and stratum sizes.** Arm B covers 6 of the 10 directions; cs-de, en-ar, en-ko, and zh-ja carry Arm C only. Sustained 8B-model generation on a fanless 8 GB machine limited the powered stratum on

some directions, so S1 is small on en-ja (4 triples) and en-ru (6). The pre-registered small-stratum guard gives point estimates only, with no CI-based claims, to any stratum with fewer than 20 matched pairs; any per-direction reading is exploratory and reported as a point estimate.

- **Edit-size imbalance.** The two Arm B directions are matched on information units, not characters: the addition changes a median of 2.0 times as many characters as the omission removes (S1 1.4x, S2 2.0x, S3 2.5x), and 79 percent of triples add more than they drop. If a metric responds to raw edit magnitude, this pushes the asymmetry *A* in the predicted negative direction, which is why the edit-magnitude sensitivity refit of Section 4 is reported alongside the primary interaction.
- **Human validation pending.** See Section 6.
- **Arm C digit scope.** Numeric pairs use Latin and Cyrillic digit contexts; native-digit corruption for zh, ja, ar, and ko scripts was not attempted.

Acknowledgments

We thank Eleftherios Avramidis and the WMT26 Automated Translation Quality Evaluation organizers for confirming the separate-segments design and the submission logistics.

References

- Chantal Amrhein, Nikita Moghe, and Liane Guillou. 2022. ACES: Translation accuracy challenge sets for evaluating machine translation metrics. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 479–513.
- Yoav Benjamini and Yosef Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46.
- Nuno M. Guerreiro, Ricardo Rei, Daan van Stigt, Luisa Coheur, Pierre Colombo, and André F. T. Martins. 2024. xCOMET: Transparent machine translation evaluation through fine-grained error detection. *Transactions of the Association for Computational Linguistics*, 12:979–995.
- Juraj Juraska, Daniel Deutsch, Mara Finkelstein, and Markus Freitag. 2024. [MetricX-24: The Google submission to the WMT 2024 metrics shared task](#). In *Proceedings of the Ninth Conference on Machine Translation*, pages 492–504, Miami, Florida, USA. Association for Computational Linguistics.
- Tom Kocmi, Eleftherios Avramidis, Rachel Bawden, Ondřej Bojar, Anton Dvorkovich, Christian Federmann, Mark Fishel, Markus Freitag, Thamme Gowda, Roman Grundkiewicz, Barry Haddow, Marzena Karpinska, Philipp Koehn, Benjamin Marie, Christof Monz, Kenton Murray, Masaaki Nagata, Martin Popel, Maja Popović, and 3 others. 2024. [Findings of the WMT24 general machine translation shared task: The LLM era is here but MT is not solved yet](#). In *Proceedings of the Ninth Conference on Machine Translation*, pages 1–46, Miami, Florida, USA. Association for Computational Linguistics.
- Tom Kocmi and Christian Federmann. 2023. Large language models are state-of-the-art evaluators of translation quality. In *Proceedings of the 24th Annual Conference of the European Association for Machine Translation*, pages 193–203.
- Klaus Krippendorff. 2018. *Content Analysis: An Introduction to Its Methodology*, 4th edition. SAGE Publications.
- Moritz Laurer, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers. 2024. [Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI](#). *Political Analysis*, 32(1):84–100.
- Alon Lavie, Greg Hanneman, Sweta Agrawal, Diptesh Kanojia, Chi-Kiu Lo, Vilém Zouhar, Frederic Blain, Chrysoula Zerva, Eleftherios Avramidis, Sourabh Deoghare, Archchana Sindhuja, Jiayi Wang, David Ifeoluwa Adelani, Brian Thompson, Tom Kocmi, Markus Freitag, and Daniel Deutsch. 2025. [Findings of the WMT25 shared task on automated translation evaluation systems: Linguistic diversity is challenging and references still help](#). In *Proceedings of the Tenth Conference on Machine Translation*, pages 436–483, Suzhou, China. Association for Computational Linguistics.
- Nikita Moghe, Arnisa Fazla, Chantal Amrhein, Tom Kocmi, Mark Steedman, Alexandra Birch, Rico Senrich, and Liane Guillou. 2025. [Machine translation meta evaluation through translation accuracy challenge sets](#). *Computational Linguistics*, 51(1):73–137.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318.
- Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the*

Tenth Workshop on Statistical Machine Translation, pages 392–395.

Ricardo Rei, José G. C. de Souza, Duarte Alves, Chrysoula Zerva, Ana C Farinha, Taisiya Glushkova, Alon Lavie, Luisa Coheur, and André F. T. Martins. 2022a. COMET-22: Unbabel-IST 2022 submission for the metrics shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585.

Ricardo Rei, Marcos Treviso, Nuno M. Guerreiro, Chrysoula Zerva, Ana C Farinha, Christine Maroti, José G. C. de Souza, Taisiya Glushkova, Duarte Alves, Luisa Coheur, Alon Lavie, and André F. T. Martins. 2022b. [CometKiwi: IST-unbabel 2022 submission for the quality estimation shared task](#). In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 634–645, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Thibault Sellam, Dipanjan Das, and Ankur Parikh. 2020. BLEURT: Learning robust metrics for text generation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7881–7892.

Matthew Snover, Bonnie Dorr, Rich Schwartz, Linnea Micciulla, and John Makhoul. 2006. A study of translation edit rate with targeted human annotation. In *Proceedings of the 7th Conference of the Association for Machine Translation in the Americas*, pages 223–231.

Yilin Zhang, Wenda Xu, Zhongtao Liu, Tetsuji Nakagawa, and Markus Freitag. 2025. [Penalizing length: Uncovering systematic bias in quality estimation metrics](#). *Preprint*, arXiv:2510.22028.

Vilém Zouhar, Wenda Xu, Parker Riley, Juraj Juraska, Mara Finkelstein, Markus Freitag, and Daniel Deutsch. 2026. [Generating difficult-to-translate texts](#). In *Proceedings of the First Workshop on Multilingual Multicultural Evaluation*, pages 204–219, Rabat, Morocco. Association for Computational Linguistics.